

JOSE PAREDES COLMENAREJO

Data Scientist | AI Engineer | ML Engineer

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PROFESSIONAL SUMMARY

Data Scientist and AI Engineer with hands-on experience building production-grade ML models, AI-assisted applications, and automation systems for industrial engineering. Architected DocFlow, a multi-tenant SaaS document management platform built using AI-assisted development (Claude/Anthropic API), orchestrating the full product lifecycle from system design to Docker deployment. Active open-source contributor with 33+ repositories, 268+ GitHub followers, and 10+ computer vision models published on HuggingFace. Strong expertise in Computer Vision (YOLO), NLP, algorithmic trading, agentic AI workflows, and end-to-end data pipeline development. Bilingual (Spanish/English).

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, Bash, HTML/CSS

AI Engineering: Anthropic Claude API, AI-assisted development, agentic workflows, context engineering, prompt design, LLM integration

Backend: FastAPI, Flask, REST APIs, JWT authentication, WebSockets, IMAP/SMTP integration

Frontend: React 18, Tailwind CSS, Framer Motion, Axios

Data Science & ML: Pandas, NumPy, scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Transformers

Computer Vision: YOLO (v9/v11/v12), OpenCV, ResNet, object detection, image classification

NLP: Sentiment analysis, text mining, Hugging Face Transformers, spaCy, ASR/speech recognition

Data & Databases: SQL, SQLite, openpyxl, Excel automation, EDA, feature engineering, time series

DevOps & Tools: Docker, Docker Compose, GitHub Actions CI/CD, Git, Jupyter, VS Code

Visualization & BI: Tableau, Matplotlib, Seaborn, Plotly

Other: PCA, K-Means, Monte Carlo simulation, Kelly Criterion, DocuSign API, multi-tenant architecture, AI agent orchestration

PROJECTS

DocFlow SaaS - Document Management Platform | [FastAPI](#), [React 18](#), [Tailwind CSS](#), [Docker](#), [JWT](#), [Anthropic API](#), [DocuSign](#)

Architected and directed AI-assisted development (Claude) of a full-stack multi-tenant SaaS for industrial document management at EIPSA. Designed system architecture, defined requirements, and orchestrated AI agents to build 21 API router modules, 24 service layers, and 24 frontend pages. Features include AI-powered document processing (Anthropic API), DocuSign e-signature integration, automated email ingestion (IMAP/SMTP), JWT authentication, i18n, backup scheduling, and Docker deployment with CI/CD via GitHub Actions. Currently in production use.

Fluorescent Penetrant Inspection Detection | [YOLO11](#), [Computer Vision](#), [HuggingFace](#), [Python](#)

Developed an AI model for automated detection of liquid penetrant test results in metallurgy quality control. Trained on industrial inspection data and deployed as a HuggingFace model for real-time defect classification in manufacturing environments.

Welding Defects Detection System | [YOLO11](#), [Computer Vision](#), [Deep Learning](#), [HuggingFace](#)

Built an object detection system for real-time identification and classification of welding defects in industrial environments. Model published on HuggingFace. 14+ GitHub stars.

DXF Data Extraction Tool | [Python](#), [CAD automation](#), [deep learning](#), [data extraction](#)

Created a tool combining deep learning with CAD parsing to automatically extract and classify entities from DXF technical drawings. Reduced manual data entry time by approximately 80% in engineering workflows. 18+ GitHub stars, 4 forks.

Polymarket Auto-Betting Bot | [Python](#), [REST APIs](#), [Kelly Criterion](#), [Monte Carlo](#), [Telegram](#)

Algorithmic trading bot for BTC prediction markets on Polymarket. Implements Kelly Criterion sizing, Monte Carlo backtesting (10,000+ simulations), bail-out logic, gasless on-chain operations, and real-time Telegram dashboard. 5+ GitHub stars.

Email MAPI Automation | [Python](#), [Outlook/IMAP](#), [Excel](#), [ERP integration](#)

Automated documentation ingestion pipeline: captures emails via Outlook/IMAP, extracts attachments and metadata, and loads structured data into Excel and ERP systems. Eliminated manual email processing for document control.

PROFESSIONAL EXPERIENCE

Data Analyst & AI Engineer

Jan 2024 - Present

EIPSA - Espanola de Instrumentacion Primaria S.A., Madrid

- Architected DocFlow, a full-stack SaaS document management platform, using AI-assisted development (Claude) to design and build the complete system (FastAPI + React). Now in production use.
- Built and deployed YOLO-based computer vision models for automated defect detection in metallurgy quality control (penetrant inspection, welding defects).
- Created automated data extraction pipelines from DXF technical drawings using deep learning, reducing manual processing time by 80%.
- Automated email-to-ERP workflows, eliminating manual document entry for 500+ monthly incoming documents.
- Built interactive Tableau dashboards for real-time project documentation tracking across international clients.
- Coordinated documentation submissions with international contractors (Técnicas Reunidas) using eGesdoc/VDDL platforms.

Data Management Technician

Jan 2021 - Dec 2023

Comunicacion Permanente S.L., Madrid

- Managed and maintained customer database with 10,000+ records, ensuring data integrity and accuracy.
- Processed and validated incoming data streams for technical incident tracking and emergency response.
- Developed internal reporting procedures that improved incident resolution tracking by 25%.

Electronics Lab Technician

Jan 2015 - Dec 2020

GRUPO ANOVO S.L., Madrid

- Diagnosed and repaired 50+ electronic devices weekly (Samsung, ZTE, multi-brand) with 95%+ quality rate.
- Managed production data and quality control metrics for device repair workflows.

EDUCATION

Data Science Bootcamp

Sep 2021 - Jun 2022

The Bridge | Digital Talent Accelerator, Madrid

- Supervised and unsupervised learning, deep learning, NLP with Transformers, Computer Vision (OpenCV), time series forecasting, data storytelling, and Flask deployment.

Advanced Vocational Degree in Telecommunications

Sep 2010 - Jun 2012

Centro de Formacion Padre Piquer, Madrid

LANGUAGES

Spanish: Native

English: Professional working proficiency (B2)

ADDITIONAL INFORMATION

Platforms: HuggingFace (jparedesDS) - 10+ published models | Kaggle (joseparedesc) | Tableau Public

Open Source: 33+ public repositories, 268+ GitHub followers, Pull Shark & Starstruck achievements

Computer Vision: 10+ YOLO models (v9-v12) for real-time object detection deployed on HuggingFace

Competitions: Kaggle Bronze Medal - Multilabel Image Classification (3rd place, ResNet152)